

Control Valve MAPS Template (Electrical)

The Control Valve MAPS template is designed to control a valve that has an analog output with an analog feedback and two end limit switches.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This document describes the Advanced Control Valve MAPS template. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

The Difference between VERSION 1 and VERSION 2 of the Templates is that now in version 2, you can disable the valve's OPEN LIMIT and CLOSE LIMIT coming from the field, and hide the status of these if selected.

Advanced Control Valve Functional Philosophy

The valve open analog output is enabled once all interlocks are ready and an open request is received.

Field Interlocks

- Safety Interlock Feedback

PLC Interlocks

- Process Interlock Ready
- There are no interlock faults latched

Interlock fault recovery process – once a field interlock is activated the face plate will display a red indication, if the interlock is recovered and still latched the faceplate will show a green indication and the reset button on the faceplate will flash, pressing on the reset button will unlatch the interlock and display a grey indication and allow the valve to become ready for operation.

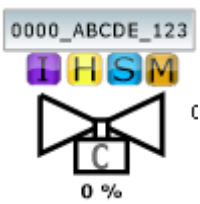
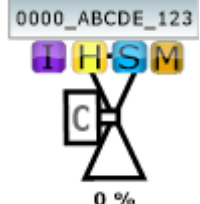
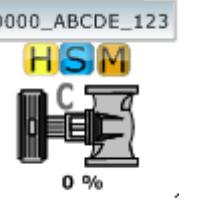
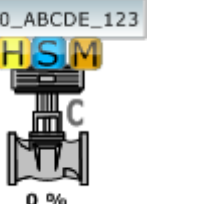
Maintenance mode - will disable the Process Interlocks, but respects all the other interlocks. This is used to test a valve while the process is closed.

Simulation mode - will disable the physical outputs of the valve and simulate an input limit feedback after the elapsed setpoint time from the faceplate. The Process Interlocks are respected. Simulation mode is used to test the process logic without opening the physical valve itself. It is very useful in commissioning and maintenance.

Open and Close requests are initiated by selecting a mode on the valve faceplate:

- **Auto mode** - the auto control program initiates an Open and Close.
- **Manual mode & Desk mode** - the operator initiates an Open and Close from the faceplate.
- **Manual mode & Field mode** - the operator initiates an Open and Close from the Open and Close push buttons in the field.

Typical Faceplate Graphics for the Advanced Control Valve MAPS Template

Valve ISO DOWN	Valve ISO RIGHT	Valve LEFT	Valve TOP
			
	Manual Mode		
	Maintenance Mode		
	Simulation Mode		
	Process Interlock		
	Valve Control / Tag Name		

Graphics legend:

Line colour	Fill colour	Valve status
Black	Grey	Closed
Grey	White	Closing
Green	White	Opening
Black	Green	Opened
Black	Yellow/Red	Fault active

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshal	x		
SSCL 1	Fail to Open Setpoint	Analog	x		
SSCL 2	Fail to Close Setpoint	Analog	x		
SSCL 3	Simulate Setpoint	Analog	x		
SSCL 4	Opened Delay Setpoint	Analog	x		
SSCL 5	Closed Delay Setpoint	Analog	x		
SSCL 6	Manual Setpoint	Analog	x		
SSCL 7	Hysteresis Setpoint	Analog	x		
SSCL 8	Minimum Raw Setpoint	Analog	x		
SSCL 9	Maximum Raw Setpoint	Analog	x		
SSCL 10	Minimum Engineering Setpoint	Analog	x		
SSCL 11	Maximum Engineering Setpoint	Analog	x		

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshal	x		
SSSL 1	Last Opened Time Process Value	Analog	x		
SSSL 2	Last Closed Time Process Value	Analog	x		
SSSL 3	Number of Operations	Analog	x		
SSSL 4	Valve Current Process Value	Analog	x		

Digital Inputs:		Advanced	Standard	Basic
DI 0	Open Limit Feedback	x		
DI 1	Closed Limit Feedback	x		
DI 2	Safety Interlock Ok	x		
DI 3	Open Push Button	x		
DI 4	Close Push Button	x		
Analog Input:				
AI 0	Valve Open Analog Status	x		
Analog Output:				
AO 0	Valve Open Analog Command	x		

SCADA Control Word (Adroit Marshal Agent) V1		Event Logged	Advanced	Standard	Basic
Bit 0	Auto Manual	x	x		
Bit 1	Field Desk	x	x		
Bit 2	Maintenance	x	x		
Bit 3	Simulation	x	x		
Bit 4	Reset Fault	x	x		
Bit 5	Reset Number of Operations	x	x		
Bit 6	Open Request	x	x		
Bit 7	Spare				
Bit 8	Close Request	x	x		
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

SCADA Control Word (Adroit Marshal Agent) V2		Event Logged	Advanced	Standard	Basic
Bit 0	Auto Manual	x	x		
Bit 1	Field Desk	x	x		
Bit 2	Maintenance	x	x		
Bit 3	Simulation	x	x		
Bit 4	Reset Fault	x	x		
Bit 5	Reset Number of Operations	x	x		
Bit 6	Open Request	x	x		
Bit 7	Spare				
Bit 8	Close Request	x	x		
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Open Limit Disabled	x	x		
Bit 15	Close Limit Disabled	x	x		

SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Process Interlock Ok		x		
Bit 1	Valve Healthy		x		
Bit 2	Closed		x		
Bit 3	Opening		x		
Bit 4	Opened		x		
Bit 5	Closing		x		
Bit 6	Fault		x		
Bit 7	Safety Fault	x	x		
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Fail to Open Alarm on Interlock	x	x		
Bit 12	Open FB		x		
Bit 13	Closed FB		x		
Bit 14	Fault Open	x	x		
Bit 15	Fault Close	x	x		

Resources	Advanced	Standard	Basic
Digital Inputs	5		
Digital Outputs	2		
Analog Inputs	1		
Analog Outputs	1		
PLC Memory Steps	651		
System Words Used	83		
System Bits Used	97		
SCADA SCAN Points	17		

Template: Advanced Control Valve PLC Function Block

<Device> (Advanced Control Valve)

	FB_VALVE_C_A_v1_0		
	VALVE_C_A_v1_0		
DATA_VALVE_C_A_V1_0.F_I_Open_FB	F_I_Opened_FB	F_O_VALVE_CONTROL	DATA_VALVE_C_A_V1_0.F_O_VALVE_CONTROL
DATA_VALVE_C_A_V1_0.F_I_Closed_FB	F_I_Closed_FB	S_SW1_Marshall	DATA_VALVE_C_A_V1_0.S_SW1_Marshall
DATA_VALVE_C_A_V1_0.F_I_Valve_Status	F_I_Valve_Status	S_SW2_Opening_Time_PV	DATA_VALVE_C_A_V1_0.S_SW2_Opened_Time_PV
DATA_VALVE_C_A_V1_0.F_I_Safety_Int_OK	F_I_Safety_Int_OK	S_SW3_Closing_Time_PV	DATA_VALVE_C_A_V1_0.S_SW3_Closed_Time_PV
DATA_VALVE_C_A_V1_0.VAR_Process_Int_OK	VAR_Process_Int_OK	S_SW4_No_Of_Operations	DATA_VALVE_C_A_V1_0.S_SW4_No_Of_Operations
DATA_VALVE_C_A_V1_0.VAR_Auto_SP	VAR_Auto_SP	S_SW5_Valve_PV	DATA_VALVE_C_A_V1_0.S_SW5_Valve_PV
DATA_VALVE_C_A_V1_0.F_I_OPEN_PB	F_I_Open_PB	S_SW1_7_Fault_Safety	
DATA_VALVE_C_A_V1_0.F_I_CLOSE_PB	F_I_Close_PB	S_SW1_E_Fault_Open	
DATA_VALVE_C_A_V1_0.S_CW1_Marshall	S_CW1_Marshall	S_SW1_F_Fault_Close	
DATA_VALVE_C_A_V1_0.S_CW2_Fail_To_Open_SP	S_CW2_Fail_To_Open_SP	S_CW1_2_Maintenance	
DATA_VALVE_C_A_V1_0.S_CW3_Fail_To_Close_SP	S_CW3_Fail_To_Close_SP	S_CW1_3_Simulation	
DATA_VALVE_C_A_V1_0.S_CW4_Simulate_SP	S_CW4_Simulate_SP	S_SW1_1_Healthy	
DATA_VALVE_C_A_V1_0.S_CW5_Opened_SP	S_CW5_Opening_SP	S_SW1_2_Closed	
DATA_VALVE_C_A_V1_0.S_CW6_Closed_SP	S_CW6_Closing_SP	S_SW1_3_Opening	
DATA_VALVE_C_A_V1_0.S_CW7_Manual_SP	S_CW7_Manual_SP	S_SW1_4_Opened	
DATA_VALVE_C_A_V1_0.S_CW8_Hysteresis_SP	S_CW8_Hysteresis_SP	S_SW1_5_Closing	
DATA_VALVE_C_A_V1_0.S_CW9_MinRaw_SP	S_CW9_MinRaw_SP	S_CW1_0_Manual_Auto	
DATA_VALVE_C_A_V1_0.S_CW10_MaxRaw_SP	S_CW10_MaxRaw_SP	S_CW1_1_Desk_Field	
DATA_VALVE_C_A_V1_0.S_CW11_MinEng_SP	S_CW11_MinEng_SP		
DATA_VALVE_C_A_V1_0.S_CW12_MaxEng_SP	S_CW12_MaxEng_SP		

Template Graphic: Advanced Home Screen

Equipment ID Valve Control

Status

0 %

Healthy

Open Limit

Close Limit

Process Interlock

Safety Fault

Open Fault

Close Fault

Control Mode

Auto Manual

Desk / Field

Desk Field

Operation

Open Close

Manual Control

Manual Position SP #####

Reset

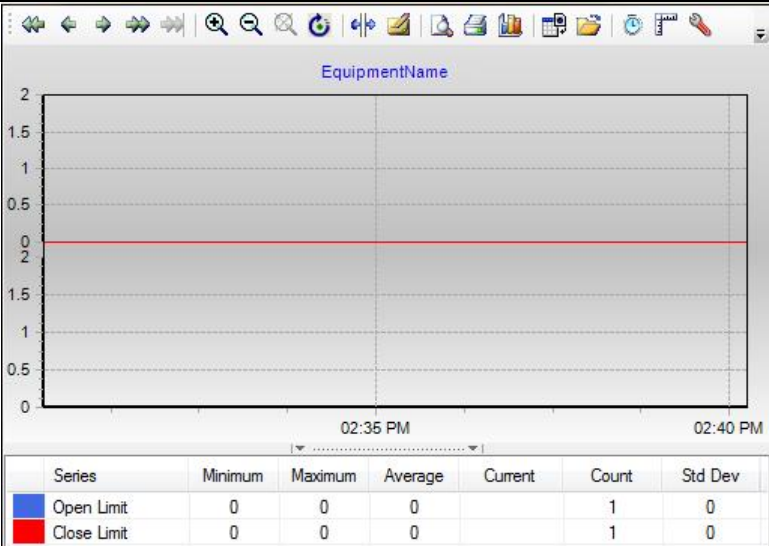
Reset

Control	Description
Auto	Auto control by PLC
Manual	Manual control by operator, desk and field mode
Desk	Manual control from the SCADA faceplate
Field	Manual control from the field Open/Close station
Open	Manual open the Valve
Close	Manual close the Valve
#####	Manual Open Position Setpoint Value
Reset	Reset the Valve after a fault has been repaired
Healthy	Healthy to Open
Open Limit	Valve opened limit input
Close Limit	Valve closed limit input
Process Interlock	Valve process ok to open interlock ready
Safety Fault	Valve safety off fault
Open Fault	Valve failed to open fault
Close Fault	Valve failed to close fault

Template Graphic: Advanced Setup Screen for Version 1	Control	Description
	Simulate	Enable simulation of Valve, disables the physical outputs
	Feedback Time	Time for simulation to simulate open/close feedback signals
	Maintenance	Enable maintenance mode (Disables all interlocks)
	Number of Operations	Total number of valve open operations
	Reset	Reset the total number of valve open operations
	Open Signal Delay (SP)	Delay the PLC to accept the valve is open
	Max Opening (SP)	Time for the valve to fail to open
	Last Opened (PV)	Last time the valve taken to open
	Close Signal Delay (SP)	Delay the PLC to accept the valve is close
	Max Closing (SP)	Time for the valve to fail to close
	Last Closed (PV)	Last time the valve taken to close
	Feedback Hysteresis	Analog feedback in range hysteresis setpoint
	MinRaw / MaxRaw	Raw minimum and maximum scaling setpoints
	MinEng / MaxEng	Engineering minimum and maximum scaling setpoints

Template Graphic: Advanced Setup Screen for Version 2	Control	Description
	Simulate	Enable simulation of Valve, disables the physical outputs
	Feedback Time	Time for simulation to simulate open/close feedback signals
	Maintenance	Enable maintenance mode (Disables all interlocks)
	Number of Operations	Total number of valve open operations
	Reset	Reset the total number of valve open operations
	Open Signal Delay (SP)	Delay the PLC to accept the valve is open
	Max Opening (SP)	Time for the valve to fail to open
	Last Opened (PV)	Last time the valve taken to open
	Close Signal Delay (SP)	Delay the PLC to accept the valve is close
	Max Closing (SP)	Time for the valve to fail to close
	Last Closed (PV)	Last time the valve taken to close
	Feedback Hysteresis	Analog feedback in range hysteresis setpoint
	MinRaw / MaxRaw	Raw minimum and maximum scaling setpoints
	MinEng / MaxEng	Engineering minimum and maximum scaling setpoints

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Description
	Equipment Name	Display the current valve name of the trend.
	Open limit	Display the status of the open limit of the valve
	Close Limit	Display the status of the close limit of the valve